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Design Homework 3: Combined Cycle (CCGT)

Assignment Background:

The goal of this assignment is to combine the optimized Brayton and Rankine cycles while still maintaining the highest possible efficiency. Slight modifications might be required to ensure the optimal cycles are still physically realistic.

Additionally, this assignment focuses on the coupling of the two cycles by designing a heat exchanger. The heat exchanger consists of a preheater, an evaporator, and a superheater, in that order.

Finally, we must consider the environmental implications and constraints when designing this combined cycle.

For the Rankine Cycle, the water is being pumped through the Potomac River. Applying the appropriate assumptions to the condenser will ensure that the environment is not being severely altered or harmed using this cycle. Ultimately, this system could be used underwater, so the temperature change from the inlet to outlet water cannot be very large. Lots of wildlife are impacted by even the smallest changes in temperature, so it is necessary to adhere to the regulations set in place by governing organizations.

For the Brayton Cycle, the environment is a very important consideration. Ships burn a lot of fuel and that has a large impact on the environment. More eco-friendly fuels are being tested, but the cost has the potential to remain high until more research is done, or the fuels become more widely available. The biggest impact with combustion is all the carbon being put into the atmosphere.

Schematics:

Below are diagrams for the Brayton Cycle, Rankine Cycle, and Heat Exchanger, respectively. The two cycles are coupled by the heat exchanger. Figure 1 shows the Brayton Cycle.

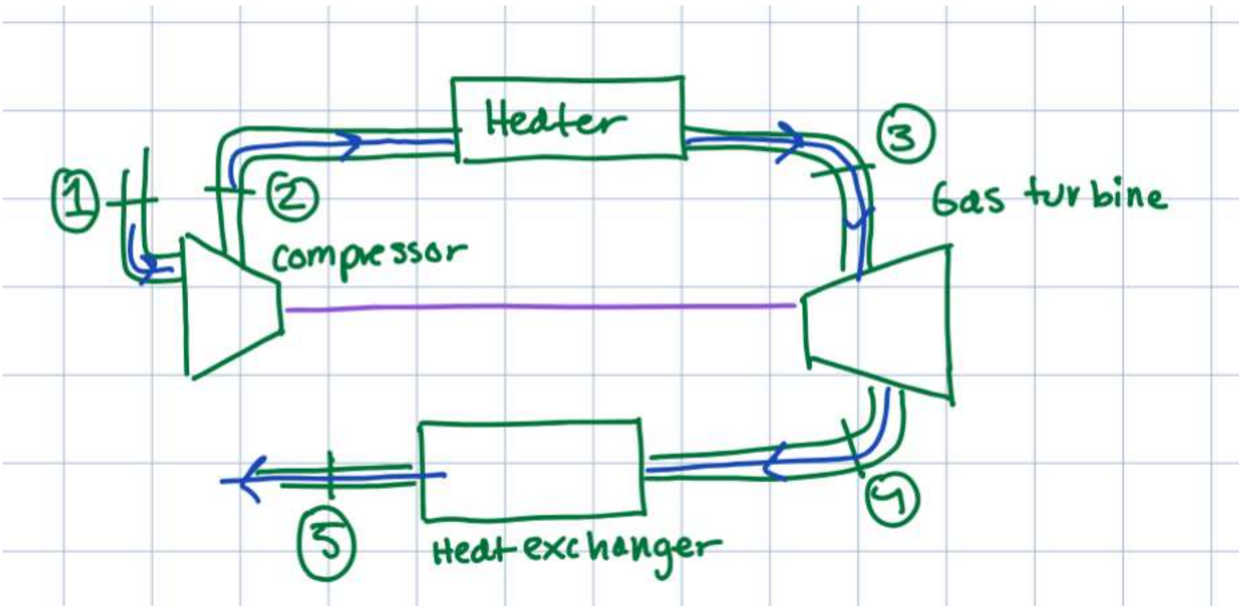

Figure. 1 Brayton (Topping) Cycle Diagram

Figure 2 shows the Rankine Cycle.

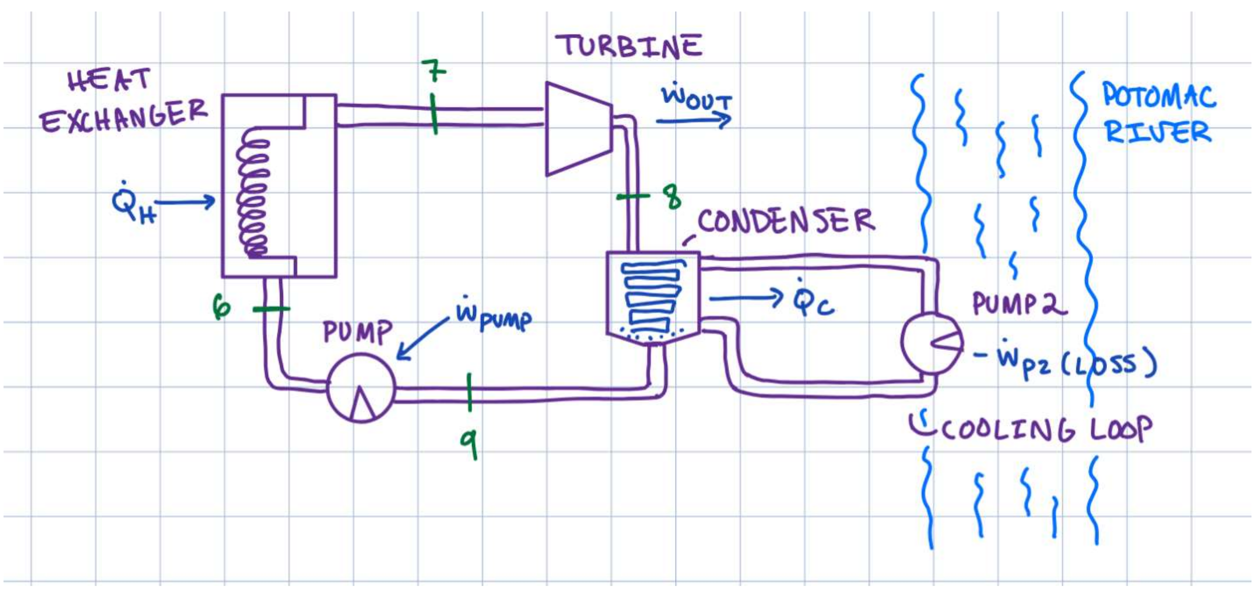

Figure. 2 Rankine (Bottoming) Cycle Diagram

Figure 3 shows the heat exchanger diagram.

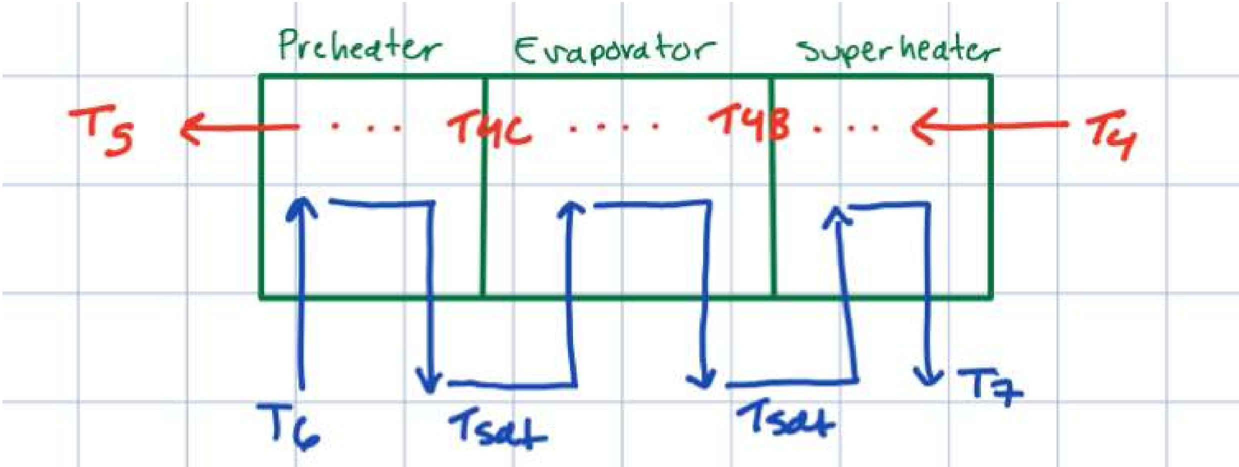

Figure. 3 Heat Exchanger Diagram. Contains a superheater, evaporator, and preheater.

T-s Diagram:

Figure 4 shows a T-s diagram for the Rankine Cycle.

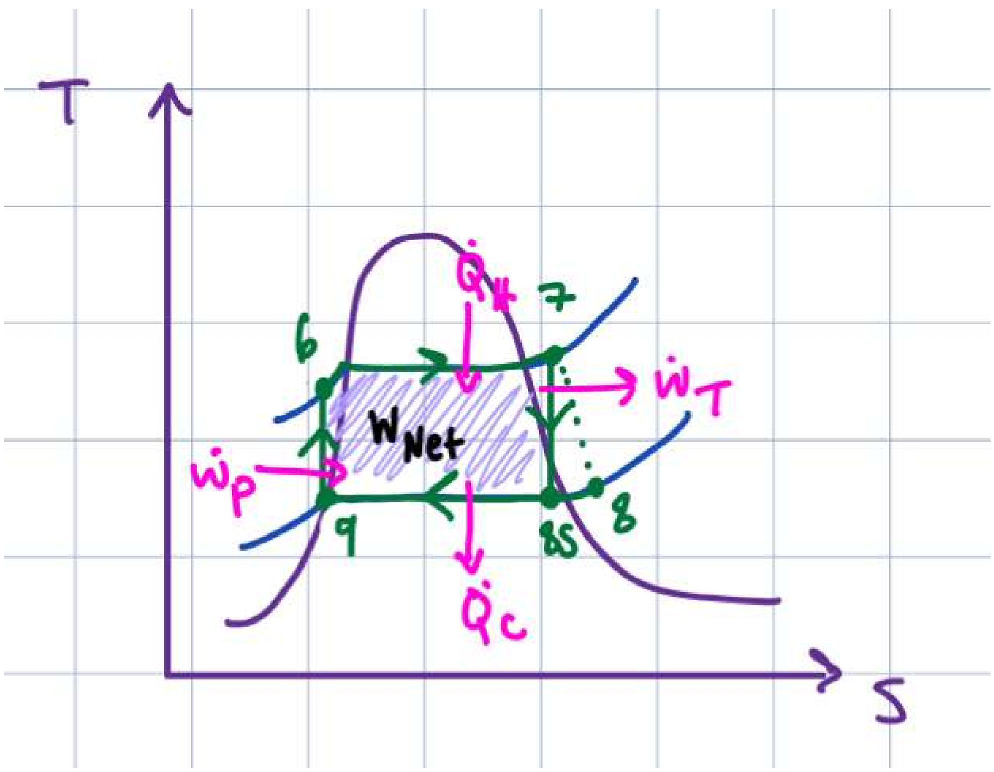

Figure. 4 T-s Diagram for the Rankine Cycle.

Figure 5 shows a T-s diagram for the Brayton Cycle.

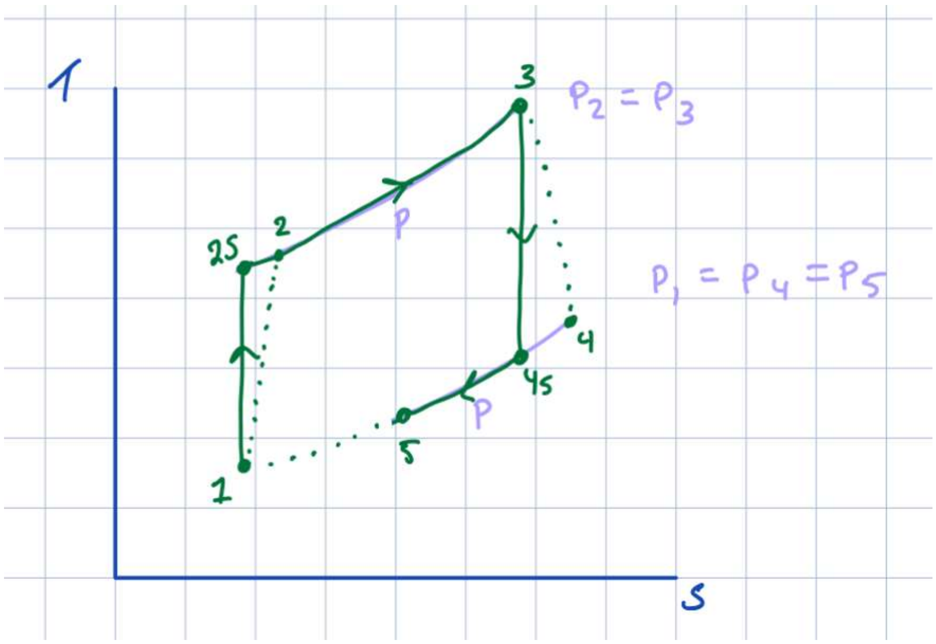

Figure 5. T-s diagram for the Brayton Cycle.

P-v Diagram:

Figure 6 shows a P-v diagram for the Rankine Cycle.

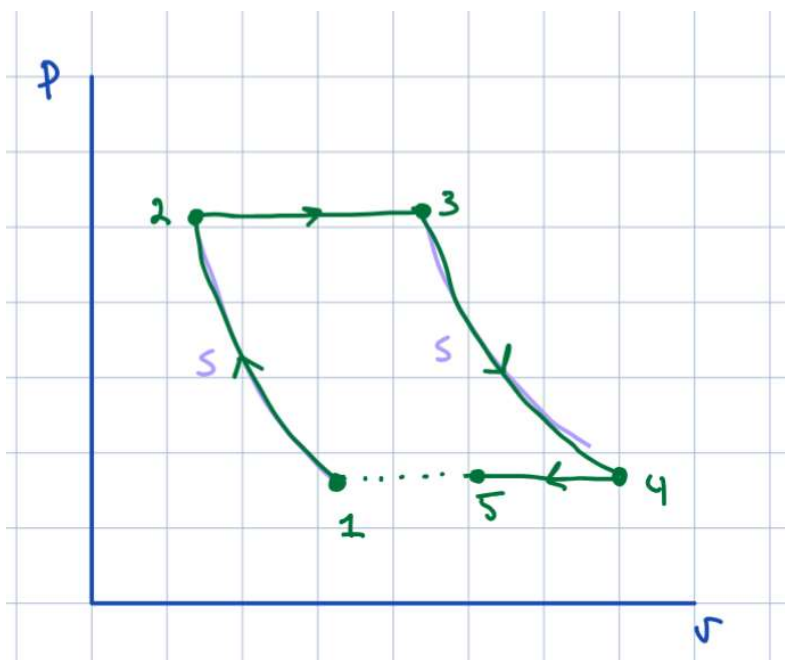

Figure 6. P-v diagram for the Rankine Cycle.

T-x Diagram:

The figures below detail the T-x diagrams for the heat exchanger. Figure 7 shows the T-x diagram for the preheater.

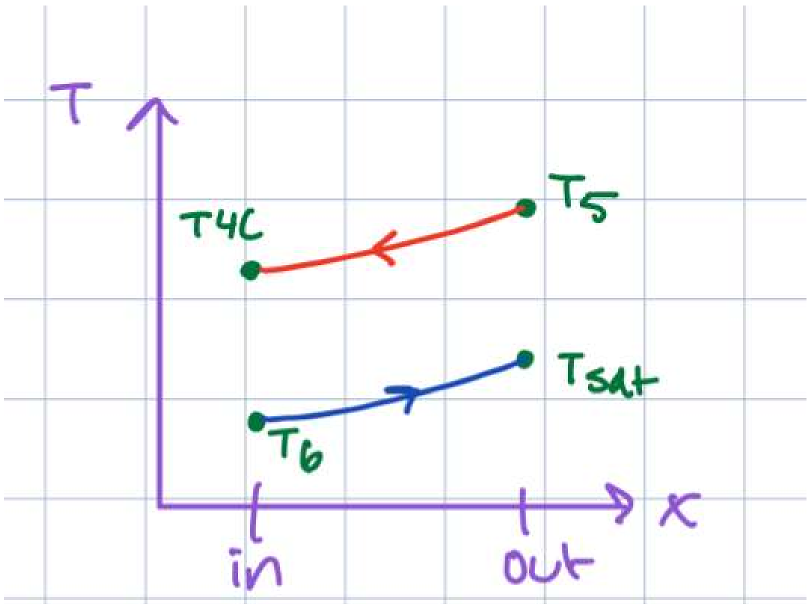

Figure 7. T-x diagram of preheater.

Figure 8 shows the T-x diagram for the evaporator.

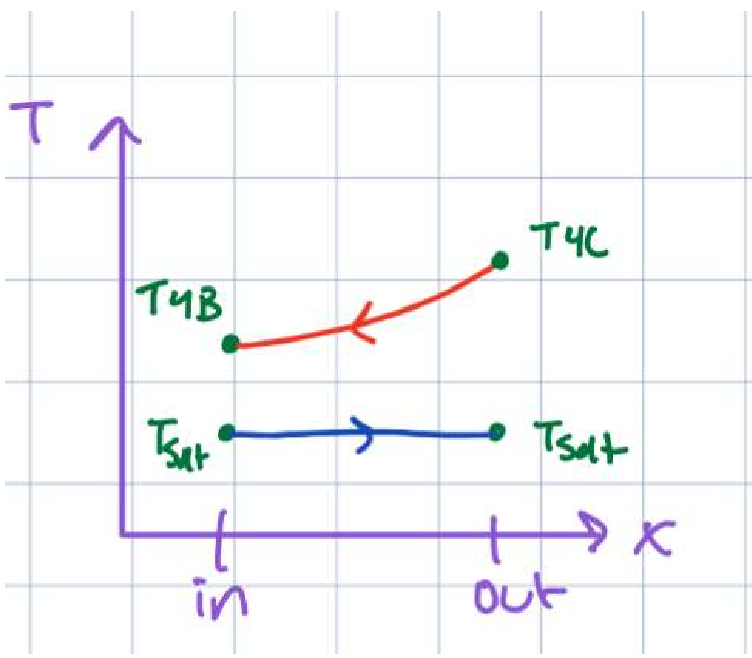

Figure 8. T-x diagram of evaporator.

Figure 9 shows the T-x diagram for the superheater.

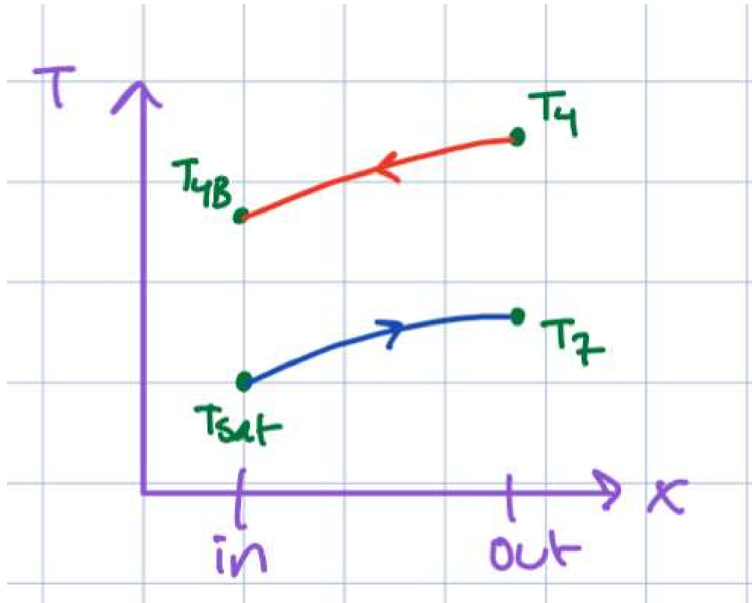

Figure 9. T-x diagram of superheater.

Assumptions and BC's:

Rankine Cycle:

General:

1) The working fluid of the cycle is **water**.
2) The coolant for the condenser is water from the **Potomac River**.
    a. The EPA has requirements on the maximum temperature rise. Ideally, the condenser will not cause more than a **1 C** change during **all** seasons (EPA). However, 3-5 C change is tolerable.
    b. The cooling water present in the condenser will be the same as the average river temperature in November. Inlet temperature will be set to **11.6 C**, while outlet temp will be set to **14.6 C** (*Eyes on the Bay*).
3) Minor losses and supply piping losses are ignored.
4) Friction losses will be considered.

5) The power needed to operate the cooling loop pump is included in efficiency calculations.
6) The fluid is incompressible. This allows use of the steam tables and simplifies calculations.
7) The system is considered ideal and closed. This is explained by sealed pipes and simplifies calculations from the real world.
8) Mass flow rate is constant throughout the system. This follows *assumption 7*.
9) Change in potential and kinetic energy is negligible. Because the system is ideal, any -changes caused by PE and KE would be very small, thus they are ignored.

Cooling Loop:

1) "The most common tube materials in a condenser are copper–nickel alloys, brass, titanium, stainless steel, and ferritic stainless steel" (Birring). For simplicity we will select stainless steel as it has a lot of data. We find that the surface roughness is around 45 micrometers (wpAdmin).

Cycle Specific:

1) States: These are the most commonly seen states in the Rankine Cycle as they maximize efficiency and minimize damage to components.
   a. State 9: Saturated Liquid
   b. State 6: Compressed Liquid
   c. State 7: Superheated Vapor
   d. State 8: 2-phase.
2) Processes 7-8 and 9-6 are isobaric.
3) P6 = P7, P8 = P9, s9 = s6, s7 = s8s, T9 = Tsat.

BCS:

1) The turbine has an isentropic efficiency of 90%
2) To minimize possible damage to turbine blades via erosion from droplet impingement, the quality should fall in the range of 80-90%, ideally falling closer to the upper end (BD Middleton).

3) The high-temperature heat exchanger ("boiler") provides 30 MW of thermal energy.
4) The pump raises the water pressure to 1000 kPa (10 bar).

Brayton Cycle:

1) The working fluid of the open Brayton cycle is air (assume IG and air standard properties) and is taken in from and exhausted into the ambient atmosphere.
2) Using the notation of the diagram in Fig. 1, the following temperatures are known: **T1 = 300 K** (environmental), **T3 = 1200 K** (material limitations), and **T5 = 500 K** (heat transfer to bottoming cycle).
3) The low-temperature heat exchanger gives off **30 MW** of thermal energy to the bottoming cycle.
4) The realistic range of compression ratios **r = p2/p1** ranges from **5 to 30**
5) The Isentropic efficiencies of the turbines and compressors will be varied independently from **70%-100%.**
6) Realistic isentropic efficiencies for the turbine and compressor are **83.5%** (Ibrahim)
7) The cycle will be using Liquified Natural Gas (LNG) which has a heat value of 50 MJ/kg (*Liquefied Natural Gas*)
8) LNG costs 7.51 $\$/ft^3$ (*Price of Liquefied U.S. Natural Gas Exports*)

Combined Cycle (CCGT):

1) The Brayton cycle operates with air and the Rankine cycle with water as working fluids, respectively.
2) The heat transfer between the two cycles is **30 MW.**
3) Both cycles are non-ideal (isentropic efficiencies for compressor and turbine of the Brayton cycle from literature search in DHW 2 and isentropic efficiency of 90% for the turbine of the Rankine cycle). The Rankine turbine outlet quality is limited (DHW 1).
4) T4 **must** be greater than T7.
5) The preheater brings the working fluid from state 6 to saturation and is a **shell-and-tube** exchanger with a **single shell** and **2 tube passes** (liquid inside tubes, gas in shell)**.**

6) The evaporator section is a **fire-tube design**, in which the hot exhaust gas from the Brayton cycle flows within the tubes and the water form the Rankine cycle boils on the outside. Assume pool boiling is occurring and ensure that the actual heat flux occurring in the evaporator is well below the critical heat flux.
7) The superheater, in which the steam of the Rankine cycle is heated from saturation to T7 (turbine inlet temperature), operates as a **counterflow** heat exchanger with two concentric tubes (steam in outer annulus, gas = air inside inner tube).
8) There are **200 thin-walled tubes** in each of the heat exchanger sections. In the preheater, they have a **diameter of 20 mm** and in the evaporator and superheater they have a **diameter of 50 mm.** The outer tube in the concentric-tube counterflow superheater has a **diameter of 80 mm.**
9) The Rankine Cycle optimized temperature will remain at 400 C. The Brayton Cycle has an optimal compression ratio of 10.26. At this ratio, the temperature, T4, is 439.9 C.
10) The rate of heat transfer in phase change is extremely high. Thus, the heat transfer coefficient for steam in the evaporator is infinity.
11) Assume thermophysical properties remain constant.
12) S_D should be no more than 1.25 – 1.5 times the diameter of the tubes (*Saari).*
13) Baffle spacing is 0.2 to 1 times the inside of the shell diameter. We will use 0.2.*(https://ijsetr.com/uploads/214536IJSETR17552-89.pdf)*

Equations:

Rankine Cycle:

The overall workflow starts at State 1, finding every value possible, going to State 2, and repeating this process through State 4. Once all values are calculated, the Work and Heat Transfer can be found. Most properties are found using the XSteam Tables.

Every equation below is a simplification of Eq. 1, the energy balance.

$$\sum E_{in} = \sum E_{out} \tag{1}$$

The work consumed by the Rankine cycle is found through Eq. 2, a simplified energy balance.

$$h_9 + w_{pump} = h_6 \tag{2}$$

Where h_1 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 9, w_{pump} $\left(\frac{kW}{kg}\right)$ is the work consumed by the Rankine cycle pump, and h_2 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 6.

The heat generated by the boiler is given in the problem statement, so once the enthalpies from states 2 and 3 are found, the specific heat generated can be used to find the mass flow rate. The specific heat is calculated in Eq. 3.

$$q_{boiler} = h_7 - h_6 \tag{3}$$

Where h_7 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 7, q_{boiler} $\left(\frac{kW}{kg}\right)$ is the work consumed by the Rankine cycle pump, and h_6 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 6.

From there, mass flow rate is calculated from Eq. 4.

$$\dot{m} = \frac{\dot{Q}_{boiler}}{q_{boiler}} = \frac{\dot{Q}_{boiler}}{h_7 - h_6} \tag{4}$$

Where $\dot{m}(\frac{kg}{s})$ is the mass flowrate, Q_{boiler} (kW) is the rate of heat transfer, q_{boiler} $(\frac{kW}{s})$ is the specific rate of heat transfer, h_7 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 7, and h_6 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 6.

To find the heat transfer out of the system through the condenser we use Eq. 5.

$$q_{condenser} = h_9 - h_8 \tag{5}$$

Where $h_8 \left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 8, found later through the isentropic turbine efficiency, $q_{condenser} \left(\frac{kW}{kg}\right)$ is the work consumed by the Rankine cycle pump, and $h_9 \left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 9.

Calculating the power output of the turbine involves a few more equations. First, the quality at the turbine outlet is found using Eq. 6.

$$x = \left(\frac{s_{8s} - s_f}{s_{fg}}\right) * 100 \tag{6}$$

Where x is the quality, $s_{8s}(\frac{kJ}{kg})$ is the ideal specific entropy and is equivalent to $s_7(\frac{kJ}{kg-K})$, $s_f(\frac{kJ}{kg-K})$ is the specific entropy of the fluid, and $s_{fg}(\frac{kJ}{kg-K})$ is the difference of the specific entropy of the fluid and the vapor.

Next, the isentropic specific enthalpy is calculated using Eq. 7.

$$h_{8s} = h_f + x * h_{fg} \tag{7}$$

Where x is the quality, $h_{8s}(\frac{kJ}{kg})$ is the ideal specific entropy, $h_f(\frac{kJ}{kg})$ is the specific entropy of the fluid, and $h_{fg}(\frac{kJ}{kg})$ is the difference of the specific entropy of the fluid and the vapor.

The isentropic power generated by the turbine can be found using Eq.8.

$$w_{turbine,s} = h_7 - h_{8s} \tag{8}$$

Where $w_{turbine,s}(\frac{kW}{kg})$ is the ideal specific turbine power output, $h_{8s}(\frac{kJ}{kg})$ is the ideal specific entropy, and h_7 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 7.

Next, the isentropic turbine efficiency is used to calculate the real specific turbine power output and the specific enthalpy at state 8 using Eq. 9. And Eq. 10.

$$w_{Turbine} = \eta_{t,s} * w_{turbine,s} \tag{9}$$

Where $w_{turbine,s}(\frac{kW}{kg})$ is the ideal specific turbine power output, $\eta_{t,s}$ is the isentropic turbine efficiency, and $w_{Turbine}(\frac{kW}{kg})$ is the real specific turbine power output.

Equation 10 calculates the specific enthalpy at state 8.

$$h_8 = h_7 - w_{Turbine} \tag{10}$$

Where $w_{Turbine}(\frac{kW}{kg})$ is the real specific turbine power output, h_7 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 7, and h_8 $\left(\frac{kJ}{kg}\right)$ is the specific enthalpy at state 8.

All these values can be multiplied by mass flowrate, but it is not necessary as units will cancel.

Cooling Loop:

To calculate the number of tubes in the condenser, the value found for the condenser heat transfer is equated to Eq. 11.

$$Q = UA\Delta T_{lim} \tag{11}$$

Where Q is the heat transfer seen in the condenser, U = h $(\frac{W}{m^2-K})$, is the heat transfer coefficient, $A = NDL\pi$ is the total surface area, and ΔT_{lim} is the log-meant temperature difference between the condensing steam in the Rankine cycle and the coolant.

ΔT_{lim} can be found from Eq. 12.

$$\Delta T_{lim} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \tag{12}$$

Where ΔT_{lim} is the log-meant temperature difference between the condensing steam in the Rankine cycle and the coolant, $\Delta T_1 = \left|T_{sat} - T_{coolant,in}\right|$, $\Delta T_2 = \left|T_{sat} - T_{coolant,out}\right|$, where T_{sat} is the saturation pressure after the condenser.

Equation 11 is rearranged to solve for the number of tubes, and we are left with Eq. 13.

$$N = \frac{Q_{condenser}}{\pi U L D \Delta T_{lim}} \tag{13}$$

Where $Q_{condenser}$ is the heat transfer rate out of the condenser, U = h ($\frac{W}{m^2-K}$), is the heat transfer coefficient, L is the length of the tube, and D is the diameter of the pipe.

The mass flowrate of the coolant is found through Eq. 14.

$$\dot{m}_{coolant} = \frac{\dot{m}}{N} \tag{14}$$

Where $\dot{m}_{coolant}$ is the mass flowrate of the coolant, m is the mass flowrate of the steam, and N is the number of tubes.

This can be used to solve for the velocity of the fluid in Eq. 15.

$$V = \frac{\dot{m}_{coolant}}{\pi r^2 \rho} \tag{15}$$

Where V is the fluid velocity of the coolant, $\dot{m}_{coolant}$ is the mass flowrate of the coolant, r is the radius of the pipe, and ρ is the density of the coolant, which in this case is water.

The velocity is then used to calculate the Reynolds number using Eq. 16.

$$Re = \frac{VD}{\nu} \tag{16}$$

Where Re is the Reynold's number, V is the velocity of the coolant, D is the pipe diameter, and ν is the kinematic viscosity of the fluid.

Because the flow was determined to be laminar, we use Eq. 17 to find the friction factor.

$$f = \frac{64}{Re} \tag{17}$$

Where f is the friction factor, and Re is the Reynold's number.

This is put into the major loss term in the Extended Bernoulli equation and can be solved for using Eq. 18.

$$F = f\left(\frac{L}{2}\right)\left(\frac{V^2}{D}\right)\rho \tag{18}$$

Where F is the loss due to friction, f is the friction factor, L is the length of the straight pipe section, V is the fluid velocity, D is the pipe diameter, and ρ is the density of the fluid.

Now, the power consumption of the pump in the coolant loop is found using Eq. 19.

$$W_{pump,coolant} = \dot{m}F \tag{19}$$

Where $W_{pump,coolant}$ is the power consumption of the pump in the coolant loop, $\dot{m}$ is the mass flowrate, and F is the loss due to friction.

Finally, the thermal efficiency of the entire cycle can be calculated using Eq. 20.

$$\eta = \frac{W_{Turbine} - W_{pump} - W_{pump,coolant}}{Q_{Boiler}} * 100\% \tag{20}$$

Where η is the thermal efficiency of the entire system, $W_{Turbine}$ is the power generated by the turbine, W_{pump} is the power consumed by the pump, $W_{pump,coolant}$ is the power required to run the coolant pump, and Q_{Boiler} is the heat generated by the boiler.

Brayton Cycle:

The following equations outline the workflow to solve for the required results and calculate the optimal results. We start at state 1 and solve all the way to state 5. Starting with Eq. 21, an isentropic relationship is used in combination with the compression ratio to calculate the ideal temperature at state 2.

$$T_{2S} = T_1 * \left(r_p\right)^{\left(\frac{k-1}{k}\right)} \tag{21}$$

Where T_{2S} [K] is the temperature at state 2 in an ideal cycle, T_1 [K] is the temperature at state 1, r_p is the compression ratio, which is varied from, 5-30 in both parts 1 and 2, and k is the ratio of the constant pressure coefficient to the constant volume coefficient.

Next, the isentropic compressor efficiency is used to calculate the actual temperature at state 2. Equation 22 highlights the relationship between the two temperatures.

$$T_2 = \left(\frac{1}{\eta_c}\right) * (T_{2S} - T_1) + T_1 \tag{22}$$

Where T_2 [K] is the real temperature at state 2 and η_c is the isentropic compressor efficiency. In part 1, this is varied in conjunction with the isentropic turbine efficiency from 70%-100% in various combinations.

Next, the specific compressor work is calculated using Eq. 23.

$$w_c = c_p(T_2 - T_1) \tag{23}$$

Where w_c [kW/kg] is the specific compressor work and c_p is the constant for a constant pressure process.

T_3 is given, so the specific heat input can be calculated using Eq. 24

$$q_H = c_p(T_3 - T_2) \tag{24}$$

Where q_H [kJ/kg] is the specific heat input in the heater and T_3 [K] is the temperature at state 3.

Next, the same process from state 1 to state 2 is used across the turbine and Eq. 25 is used to calculate the ideal temperature at state 4.

$$T_{4s} = T_3 \left(\frac{1}{r_p}\right)^{\left(\frac{k-1}{k}\right)} \tag{25}$$

Where T_{4s} [K] is the ideal temperature at state 4.

The real temperature at state 4 is calculated using Eq. 26.

$$T_4 = T_3 - \eta_t(T_3 - T_{4s}) \tag{26}$$

Where η_t is the isentropic efficiency.

Next, the specific turbine work is calculated using Eq. 27.

$$w_t = c_p(T_3 - T_4) \tag{27}$$

Where w_T [kJ/kg] is the specific turbine work.

Using the given temperature at state 5 and Eq. 28, the specific heat output can be calculated. This will be used later to calculate the mass flowrate of the system.

$$q_L = c_p(T_4 - T_5) \tag{28}$$

Where q_L [kJ/kg] is the specific heat output of the heat exchanger and T_5 [K] is the temperature given at state 5.

Finally, the mass flowrate, efficiency, and net power output can be calculated for part 1. Using Eq. 29, the mass flowrate is calculated.

$$\dot{m} = \frac{Q_L}{q_L} \tag{29}$$

Where Q_L [kW] is the heat output of the heat exchanger and m [kg/s] is the mass flowrate.

Using Eq. 30, the thermal efficiency of the system is calculated.

$$\eta_{th} = \frac{W_t - W_c}{Q_H} \tag{30}$$

Where Q_H [kW] is the heat input to the system, W_c [kW] is the compressor work of the system, and W_t [kW] is the turbine work of the system.

Using Eq. 31, the net power output of the system is calculated.

$$W_{net} = W_t - W_c \tag{31}$$

Where W_c [kW] is the compressor work of the system, and W_t [kW] is the turbine work of the system.

The non-specific values are obtained by multiplying the specific values with the mass flow rate of the system.

The same exact process is used to calculate the optimal compression ratio, the only difference is the isentropic efficiencies are no longer varied, instead they are found from literature and stated in the assumptions.

To calculate the cost of operation of a marine propulsion system using liquid natural gas, the following process is used. Starting with Eq. 32, we solve for the fuel consumed.

$$fuelUsed = \frac{Q_{H,optimal}}{heatRate} \tag{32}$$

To obtain the cost of operation over time, we use Eq. 33.

$$Cost\ per\ Time = unitCost * fuelUsed \tag{33}$$

Heat Exchanger:

All the heat exchangers follow the same general process to calculate intermediate temperature and heat transfer. Equations 34 and 35 solve for each.

$$Q = m_s(h_{out} - h_{in}) \tag{34}$$

$$Q = m_a C_{p,air}(T_{in} - T_{out}) \tag{35}$$

Where Q is the heat transfer, m is the mass transfer rate for both steam and air, C_p is the constant for a constant pressure process, h is the specific enthalpy, and T is the temperature at the inlet or outlet.

<u>*Design Equations:*</u>

The same general process is followed for each section of the heat exchanger. Find the Reynolds number, calculate the Nusselt number from the Nu-Re correlation, calculate the heat transfer coefficient from the Nusselt number, find the total heat transfer coefficient, and finally, use either the $\varepsilon - NTU$ method or the log-mean method.

Superheater:

The superheater is made up of 200 thin-walled concentric tubes. The steam flows through the annulus and the air flows through the inner tube. We use the log-mean method. The Reynolds number of steam is calculated using Eq. 36.

$$Re = \frac{D_h m}{\mu A N} = \frac{(D_o - D_i)m}{\mu N \left(\frac{\pi}{4}\right)(D_o^2 - D_i^2)} \tag{36}$$

Where D_h [m] is the hydraulic diameter, m [kg/s] is the mass flow rate, μ is the dynamic viscosity, N is the number of tubes, and D_o and D_i [m] are the outer and inner diameters, respectively. The Reynolds number is calculated in this way because of the annular shape that the steam flows inside of.

The Reynolds number of air is calculated using Eq. 37.

$$Re = \frac{4m}{\mu \pi N D_i} \tag{37}$$

Where , m [kg/s] is the mass flow rate, μ is the dynamic viscosity, N is the number of tubes, and D_i [m] is the outer and inner diameter.

Because the flows are turbulent (>10,000 Re), we use Eq. 38 to find the Nusselt number.

$$Nu = 0.023Re^{\frac{4}{5}}(\mathrm{Pr})^n \tag{38}$$

Where Re is the Reynolds number, Pr is the Prandtl number, and n is either 0.3 or 0.4, depending on whether the fluid is being cooled or heated, respectively.

Next, the heat transfer coefficient is calculated using Eq. 39.

$$h = \frac{Nu * k}{D} \tag{39}$$

Where Nu is the Nusselt number, D is the diameter, or hydraulic diameter [m], and k is the thermal conductivity.

The total heat transfer coefficient is then calculated using Eq. 40.

$$U = \left[\frac{1}{h_{air}} + \frac{1}{h_{steam}}\right]^{-1} \tag{40}$$

Where h [W/m^2-K] is the heat transfer coefficient of either steam or air. In the evaporator, $h_{steam} = \infty$.

We calculate the log-mean temperature using Eq. 41.

$$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)} \tag{41}$$

Where $\Delta T_2 = T_{h,o} - T_{c,o}$ and $\Delta T_2 = T_{h,i} - T_{c,i}$.

Finally, we rearrange Eq. 42 to get Eq. 43, where we solve for the necessary length of the tubes.

$$Q = UA\Delta T_{lm} \tag{42}$$

Where U is the total heat transfer coefficient, A is the surface area over which heat transfer occurs, and T_{lm} is the log-mean temperature difference.

$$L = \frac{Q}{U\Delta T_{lm}(ND)\left(\frac{\pi}{4}\right)} \tag{43}$$

Where Q is the heat transfer, U is the total heat transfer coefficient, D is the diameter of the surface over which heat transfer occurs, the inner diameter in the case of the superheater, and N is the number of tubes.

Evaporator:

The same process is repeated for the Evaporator. In addition, we calculate the heat flux using Eq. 44.

$$q'' = \frac{Q}{A_s} \tag{44}$$

Where Q is the heat transfer and A_s is the surface area over which heat transfer occurs.

We then compare it to the critical heat flux obtained using Eq. 45

$$q''_{CHF} = C_Z h_{fg} \rho_v \left(\frac{\sigma_{lv} g (\rho_l - \rho_v)}{\rho_v^2} \right) \tag{45}$$

Where C_z is the Zuber constant and equals 0.131, h is the enthalpy difference between liquid and gas, ρ is the density of the fluid and the gas, and σ is the surface tension. σ is found using the Laplace Equation:

$$\Delta \rho_i = \rho_v - \rho_l = \frac{2\sigma_{lv}}{R} \tag{46}$$

Where ρ is the density of the fluid and the gas, σ is the surface tension, and R is the radius of the bubble and can be found using Eq. 47.

$$R(t) = \left(\frac{2T_l - T_{sat}(\rho_l) h_{fg} \rho_v}{3T_{sat}(\rho_l) * \rho_l} \right)^{\frac{1}{2}} t \tag{47}$$

Where T is the temperature of the liquid and the saturation temperature, t is time, h is enthalpy, and ρ is density. However, surface tension of water can also be found in table A.6. Thus, equations 46 and 47 are not necessary for this calculation.

Preheater:

The Reynolds number is calculated using Eq. 48.

$$Re = \frac{D_e G_s}{\mu_s} \tag{48}$$

With shell-side mass flow rate, where m is mass flow rate, and S_m is the shell-side effective cross flow area. Equation 49 gives:

$$G_s = \frac{m_s}{S_m} \tag{49}$$

With equivalent diameter,

$$D_e = \frac{4\left(C_{p,d}S_D^2 - \frac{\pi d^2}{4}\right)}{\pi d} \tag{50}$$

Where S_D is the diagonal spacing between staggered tubes and $C_{p,d}$ is equal to 0.86 for the triangular pitch of the shell.

Equation 51 gives us the shell-side effective cross flow area,

$$S_m = (baffle\ spacing) * (S_D - D) \tag{51}$$

Where D is the diameter and baffle spacing is 0.2.

For a lower Reynolds number, we use the Kern method, seen in Eq. 52.

$$Nu = 0.36Re_s^{0.55}(Pr_s)^{\frac{1}{3}} \tag{52}$$

Where Re is the Reynolds number and Pr is the Prandtl number. After these numbers are calculated, the same workflow as the superheater and evaporator are used. We calculate the heat transfer coefficient of air and steam using Eq. 39, and the overall heat transfer coefficient using Eq. 40. The log-mean temperature method is used again, so we repeat the same process

Results:

Rankine Cycle:

Table 1. Results – Final optimized results for the Rankine Cycle.

Result	**Value**	**Units**
Rankine Cycle Mass Flow Rate	9.62	kg/s
Turbine Output Power	8.44	MW
Pump Power Consumption	9.49	kW
Coolant Pump Power Consumption	20.33	kW
Optimal Turbine Inlet Temp	400	Celsius
Turbine Exit Quality	88	%
Thermodynamic Efficiency of Cycle	28.5	%
Number of Tubes in Condenser	1,280	#

Figure 10 depicts the effects of increasing turbine inlet temperature on the thermal efficiency of the cycle.

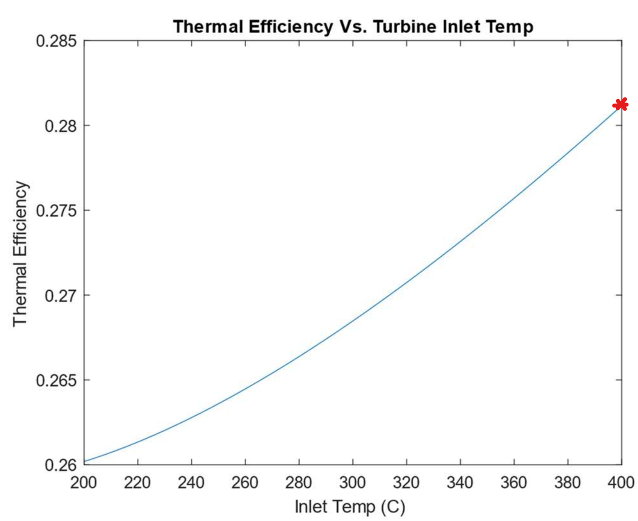

Fig. 10 Thermal Efficiency Vs. Temperature

The lower temperature bound of 200 C is chosen through optimization. Figure 11 shows behavior prior to 200 C.

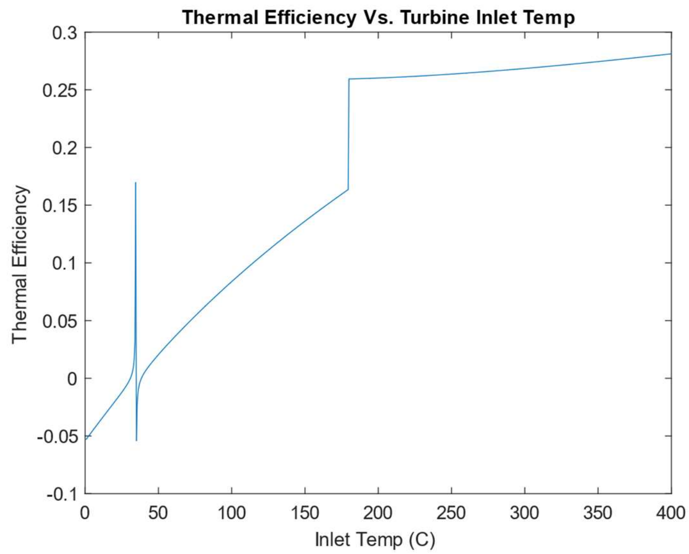

Fig. 11. Full Temperature Range

As can be seen in Fig. 11, around 200 C there is a discontinuity, and efficiency is around 0.15. This indicates that inlet temperatures should be as high as possible, and 200 C is a reasonable cut off.

Brayton Cycle:

The following combinations of isentropic turbine and compressor efficiencies are detailed in Table 2.

Table 2. Efficiencies – Combinations used to determine optimization recommendation.

Isentropic Turbine Efficiencies (%)	Isentropic Compressor Efficiencies (%)
100	100
90	100
80	100
70	100
100	90
100	80

Isentropic Turbine Efficiencies (%)	Isentropic Compressor Efficiencies (%)
100	70
90	90
80	80
70	70

Next, the ideal compression ratio is calculated for each combination of isentropic efficiencies and is shown in Table 3.

Table 3. Optimal Compression Ratios – Optimal Ratios of Compression for each combination.

Turbine & Compressor Combination (%)	Optimal Compression Ratio for Efficiency	Efficiency @ Optimal Ratio
100-100	30	0.621
90-100	26	0.440
80-100	14	0.303
70-100	9	0.203
100-90	30	0.563
100-80	23	0.467
100-70	13	0.367
90-90	18	0.380
80-80	8	0.211
70-70	5	0.091

The efficiencies of each combination are plotted versus the compression ratio range to determine which component is best optimized in the system. Figure 10 depicts the trends exhibited through increasing compression ratio and changing the combination of isentropic component efficiencies.

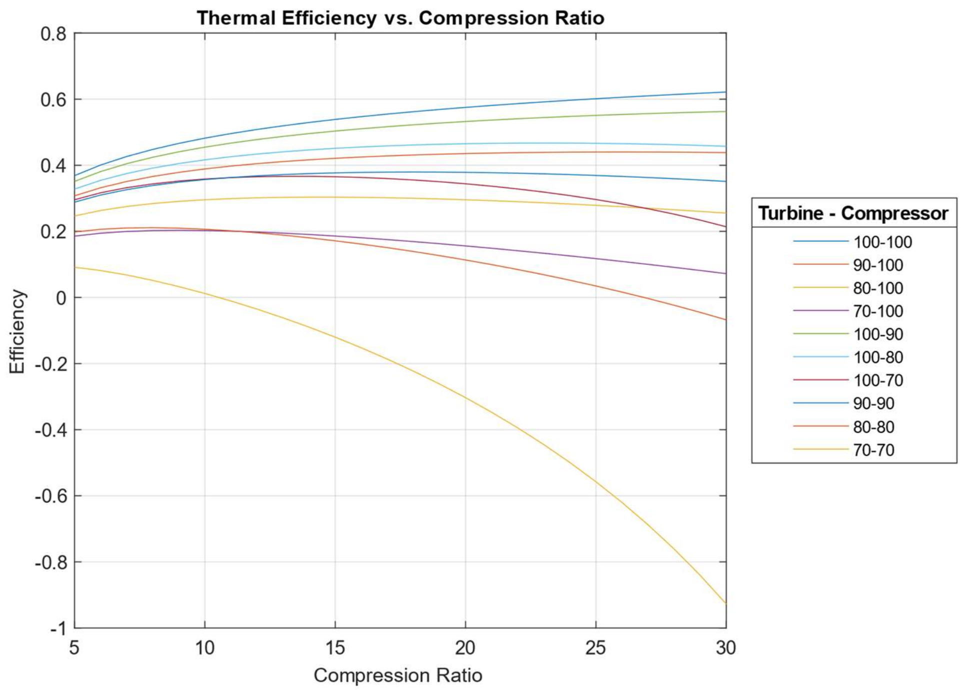

Figure. 12. Thermal Efficiency Vs. Compression Ratio for 10 Different Combinations

As can be seen in Fig. 12 and Table 3, 100% efficiency in both components is the ideal scenario, however, this is not possible. Instead, the part that provides the biggest boost to the efficiency is selected for optimization. To make this recommendation the 70-100 and 100-70 combinations are compared to the 70-70 combination. Whichever line sits higher on the graph in Fig. 2 and has the higher efficiency in Table. 2 determines which component should be prioritized. We can see that when the turbine efficiency is raised to 100% and the compressor remains at 70%, not only is the line on the graph higher than the reverse scenario, but it also has a higher efficiency. Increasing the turbine efficiency raises the system efficiency to 36.7% from 9.1%, while increasing the compressor efficiency only raises the system efficiency to 20.3%. Thus, it is recommended to optimize the turbine for best performance.

Table 4 details all the results from the optimization and the implications on the marine propulsion system, such as how much fuel is required and how much it costs to burn the specified amount of fuel.

Table.4. Final Answers – Final Values from the Optimal Compression Ratio with Realistic Isentropic Efficiencies

Final Answer	Value	Units
Optimal Compression Ratio	10.26	N/A
Optimal Thermodynamic Efficiency	26.28	%
Mass Flowrate	140.18	kg/s
Net Power Output	20.73	MW
Required Heat Input	78.88	MW
Temperature @ Optimal Compression Ratio	439	Celsius
Amount of Fuel	1.57	kg/s
Cost of Fuel	0.41	$/sec

The optimal compression ratio is identified in Fig. 11.

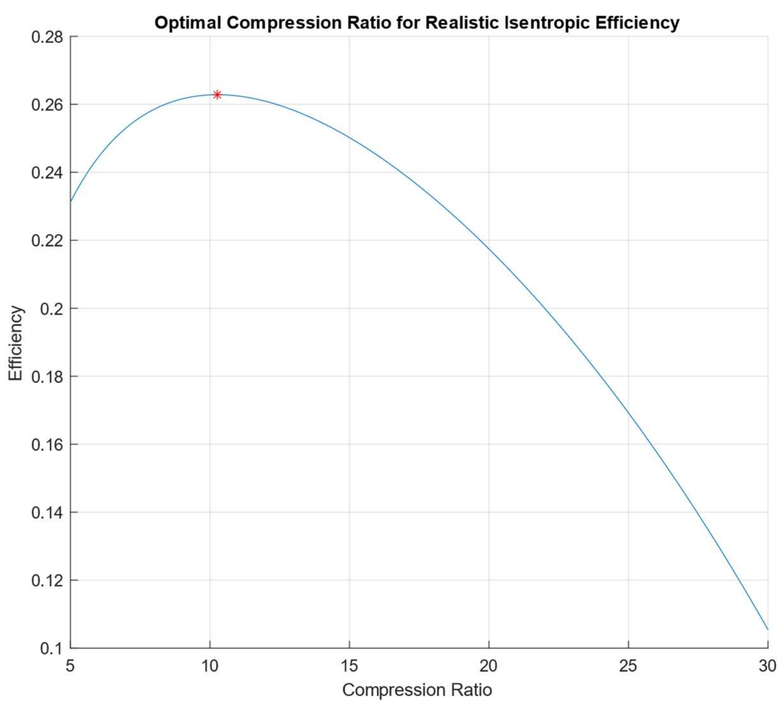

Figure. 13. Optimal Compression Ratio using Realistic Isentropic Efficiencies

It can be seen in Fig. 13, that the efficiency peaks around the lower compression ratios and really drops off when they get too high. This is due to the way the cycle is set up. This trend is furthered by plotting the net power output against the compression ratios in Fig. 12.

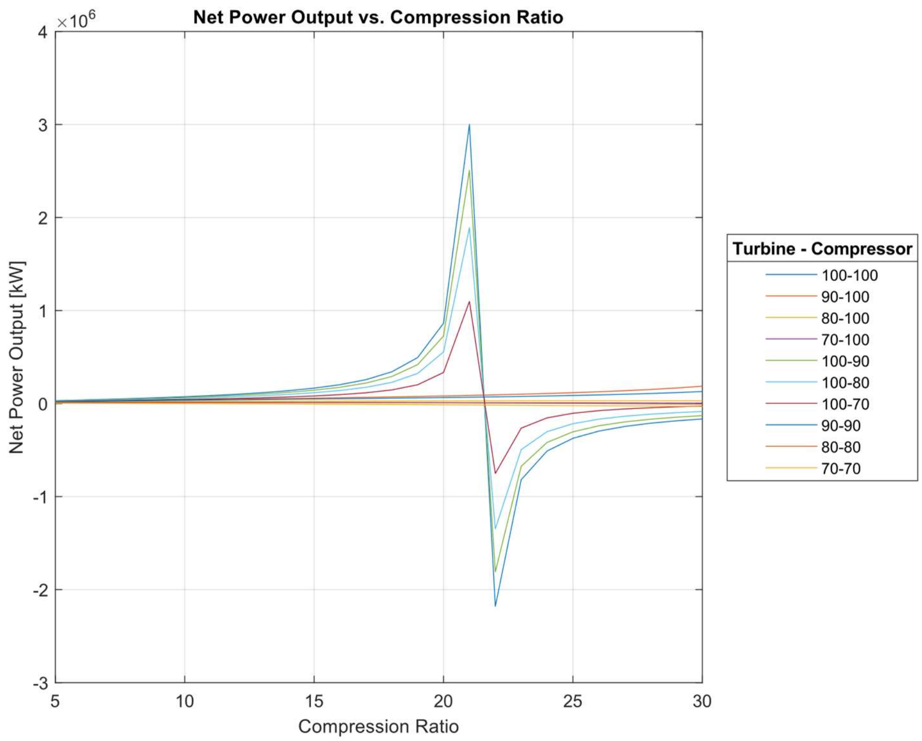

Figure. 14. Net Power Output Vs. Compression Ratio for Various Efficiency Combinations

It can be seen in Fig. 14 that there is an asymptote between a compression ratio of 21-23 and then the power output becomes negative. This indicates that past that compression ratio, the cycle is not physically possible. This occurs because of the fixed temperatures at states 1, 3, and 5. Because these temperatures are fixed, when the compression ratio increases, the temperature at state 2 increases and the temperature at state 4 decreases. When the compression ratio increases by a lot, this can lead to temperature 2 increasing above temperature 3 and temperature 4 dropping below temperature 5. Again, this is not physically possible. In the case of temperature 4 dropping below temperature 5, that would indicate heat transfer entering the system, but we know this is not the case and this is not possible. The heat transfer is supposed to be leaving the system at this state. This behavior is expected because of the constraints on the system.

Additionally, we know that some things are impossible. The cycle still must obey basic laws of thermodynamics, and too high of a compression ratio would cause those to be broken.

CCGT:

Table 5. – Results: Final results of the combined cycle.

Final Answer	Value	Units
Superheater Air Inlet Temp (T4)	439	Celsius
Superheater Air Exit Temp (T4B)	391.08	Celsius
Evaporator Air Inlet Temp (T4B)	391.08	Celsius
Evaporator Air Exit Temp (T4C)	225.6	Celsius
Preheater Air Inlet Temp (T4C)	225.6	Celsius
Preheater Air Exit Temp (T5')	225.53	Celsius
Superheater Heat Transfer (Q)	6731	W
Evaporator Heat Transfer (Q)	23,270.59	W
Preheater Heat Transfer (Q)	9.427	W
Superheater Tube Length	13.6	m
Evaporator Tube Length	8.8	m
Preheater Tube Length	1.61	m
Evaporator Heat Flux	84.47	W/m^2
Critical Heat Flux	4.645	kW/m^2
Overall Net Work	29.144	MW
Overall Efficiency	37	%
Baffle Spacing	0.2	
S_D	25	mm
S_m	0.001	mm

Sanity Check:

Rankine :

As given in the initial problem statement, a Combined Cycle Gas Turbine can have an efficiency of up to about 60%. The maximum efficiency seen here is about 28.5%. This value is very close to 30%. This makes sense because the Rankine cycle is half of the total cycle. Additionally, from the assumptions and literature review, it was determined that the quality at the turbine outlet should be somewhere from 80-90%, ideally on the upper range (Bd Middleton). The optimization found the final quality to be ~88%. This value is very close to the recommended standards and thus confirms validity.

Additionally, for the friction factor, the surface roughness found for the tubes is very low. This indicates that the pipes will behave very similar to smooth pipes and not have a huge effect on the flow. Considering the heat exchanger needs to bring the temperature down from 400 C to 34.6 C, a lot of heat needs to be taken out. When we toured the power plant on campus even the small boiler seemed to have around 20-40 tubes. For a cooling operation of this magnitude, above 1000 tubes do not seem unreasonable.

The only hiccup in these results is the Reynold's number. I would expect the pipe flow to be turbulent, however it was found to be very laminar.

Brayton:

There is not a great way to check some of the cycle parameters, but we can check the validity slightly alongside a check against how much fuel a ship typically consumes each day. A container ship cruising normally will go through around 150 tons of fuel traveling at 21 knots (*Fuel Consumption by Containership Size and Speed*). Assuming our marine propulsion system is desired to operate at the same level we can calculate how much fuel is consumed in a day and compare the numbers. Burning 1.57 kg of fuel every second means each day the ship will go through around 149 tons. 149 is extremely close to 150, thus we can assume the calculations are valid.

Heat Exchangers:

As stated in class, any length in between 2-20 meters is reasonable for the length of the tube in the heat exchanger. All the lengths are either in this range or very close (~1.6, ~8.7, ~13.6). Additionally, we are given that the total heat transfer is 30 MW and T5 is 500K. When calculating the intermediate temperatures and heat transfers all the way to T5, we find that the heat transfer in each section adds up to 30 MW and the found T5 value is only 1 degree off of the constrained value, so these check out. Also, our heat flux is well below the critical heat flux, which is desired and makes sense. However, one thing to keep in mind is the equations used to calculate Reynolds and Nusselt numbers. The equations used work for turbulent flow, but they are the simplest versions, and will not work as well as the Re gets higher and higher. This can cause some inaccuracy in results.

CCGT:

In the Design Homework statement, we are told that CCGT cycles can get up to ~60% efficiency, but ours might be a good deal lower given the design constraints. This is certainly the case. We find a final efficiency of about 37%, which is much better than the sub 30% efficiencies for the cycles on their own, but definitely falls short of the ideal goal. In order to ensure operation of the whole entire cycle, we must have a "pinch point," and the optimized states do. There is around a 40 degree difference between the temperature at state 4 and the temperature at state 7. Because the Brayton cycle was originally optimized for efficiency and the isentropic efficiencies were in the low 80s, the cycle generates more heat, thus we do not need to re-optimize the cycle as the pinch point already exists.

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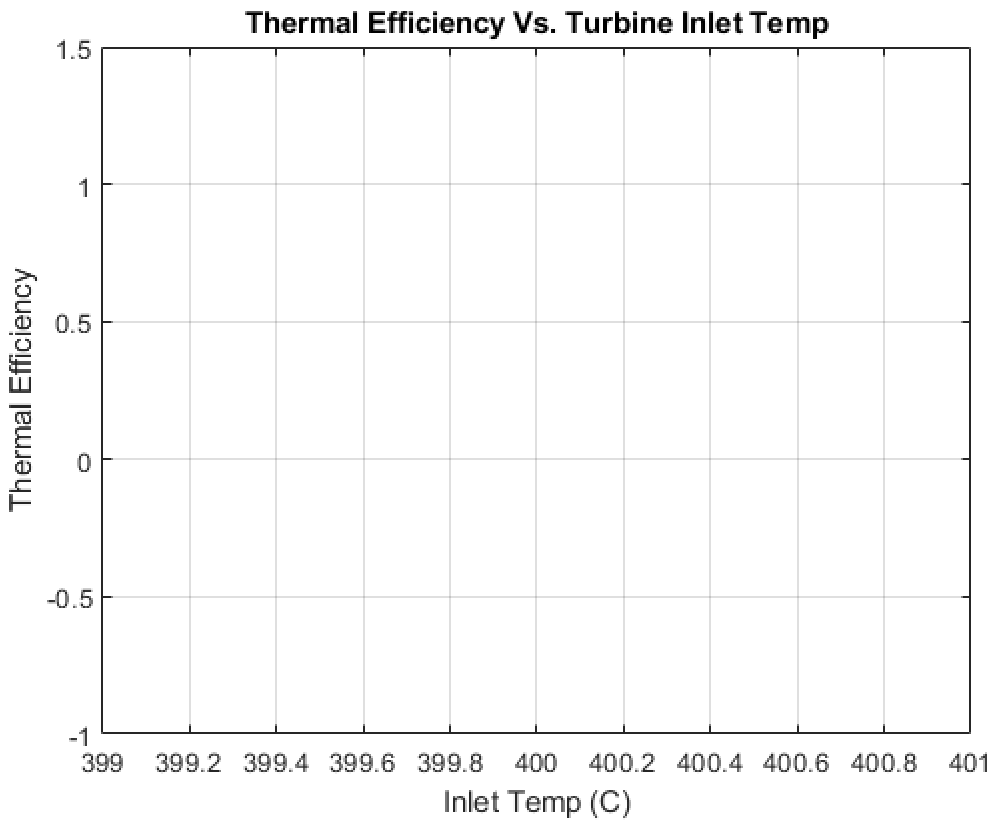
```
% Clean up the Command Window
clear;
close all;
clc;

T3 = 400;
n = zeros(1,length(T3));

for z = 1:length(T3)

% Set up Boundary Conditions

% Temperature Bounds:

T_Cond_in = 11.6;              % Coolant Inlet Temperature (Celcius)
T_Cond_out = T_Cond_in + 3;   % Coolant Outlet Temperature (Celcius)

% Givens:

Q_Boiler = 30E3;               % Thermal Energy in Through Boiler (MW)


% State 1
T1 = T_Cond_out + 20;          % Saturation Temperature (Celcius)
s1 = XSteam('sL_T',T1);        % Specific Entropy
v1 = XSteam('vL_T',T1);        % Specific Volume
h1 = XSteam('hL_T',T1);        % Specific Enthalpy
p1 = XSteam('psat_T',T1);

% State 2
s2 = s1;                       % Specific Entropy (Isentropic)
p2 = 10;                       % Pressure(Bars) @ State 2: Heat Exchanger
T2 = XSteam('T_ps',p2,s2);    % State 2 Temp (Celcius)
v2 = v1;
h2 = XSteam('h_ps',p2,s2);

% State 3
p3 = p2;                       % Pressure(Bars) @ State 3: Turbine
h3 = XSteam('h_pT',p3,T3(z));        % Specific Enthalpy
s3 = XSteam('s_pT',p3,T3(z));        % Specific Entropy

% State 4
s4s = s3;                      % Specific Entropy (Isentropic)
T4 = XSteam('Tsat_p',p1);            % Boundary Condition
sf = XSteam('sL_T',T4);
sfg = XSteam('sV_T',T4) - sf;
x = (s4s-sf)/sfg;              % Calculate Quality to get isentropic enthalpy
hf = XSteam('hL_T',T4);
hfg = XSteam('hV_T',T4) - hf;
h4s = hf + x*hfg;              % Calculate Isentropic Enthalpy

%Turbine
```

```
isoTurbineEff = 0.9;
wTs = h3 - h4s;
wT = isoTurbineEff*wTs;
h4 = h3 - wT;

%Boiler
q_Boiler = h3-h2;
m = Q_Boiler/q_Boiler;

%Condenser
q_Condenser = h1-h4;

%Pump
wPump = h2 - h1;

% Coolant Loop

% Number of Tubes

L = 10;
D = 25E-3;
h = 1000;
rho = 999;
v = 1.12E-6;
eps = 45E-6;
delT1 = T4 - T_Cond_in;
delT2 = T4 - T_Cond_out;

Tlim = (delT1-delT2)/log(delT1/delT2);

N = (-m*q_Condenser*10^3)/(h*pi*L*D*Tlim);

mCoolant = m/N;

vCoolant = mCoolant/(pi*0.25*D*rho);

Re = vCoolant*D/v;

%f = (-1.8*log10((((eps/D)/3.7))^1.11 + 6.9/Re))^-1;
f = 64/Re;

FrictLoss = (f*rho*L*vCoolant^2)/(2*D);

wPump2 = m*FrictLoss;

% Overall Efficiency

sumWork = m*(wT - wPump)- wPump2;

n(z) = sumWork/Q_Boiler;

end

plot(T3, n);
```

```
title('Thermal Efficiency Vs. Turbine Inlet Temp');
xlabel('Inlet Temp (C)');
ylabel('Thermal Efficiency');
grid on
```



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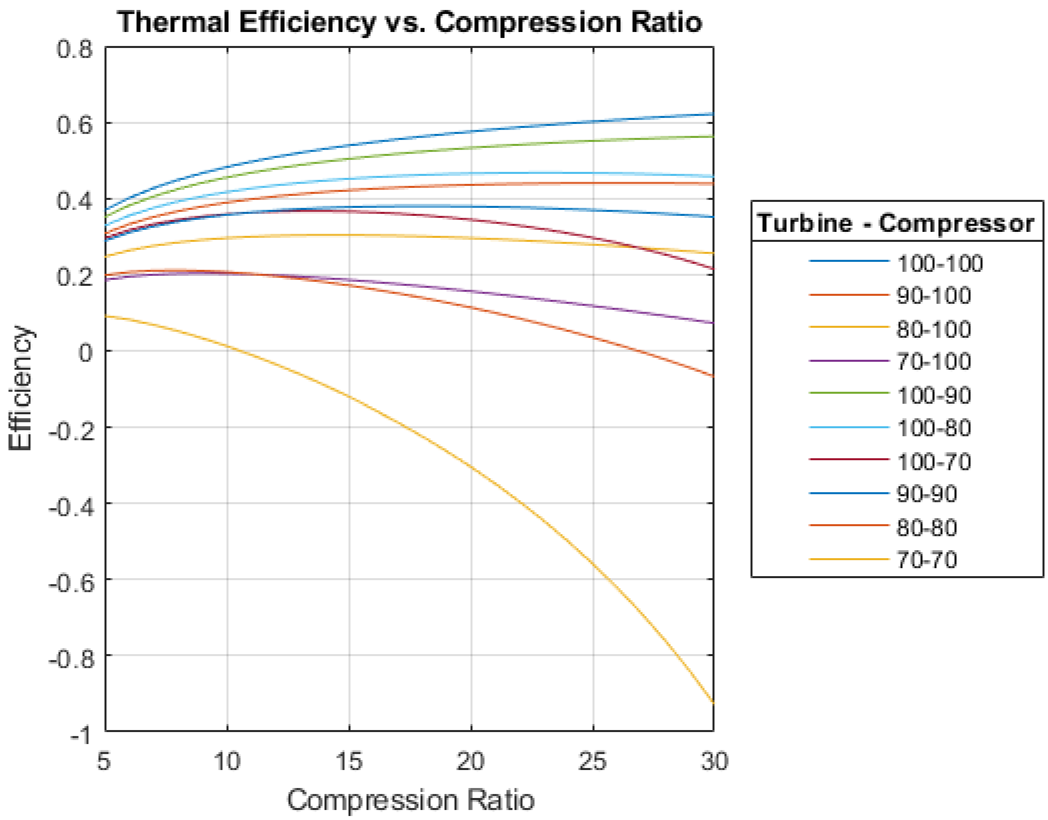
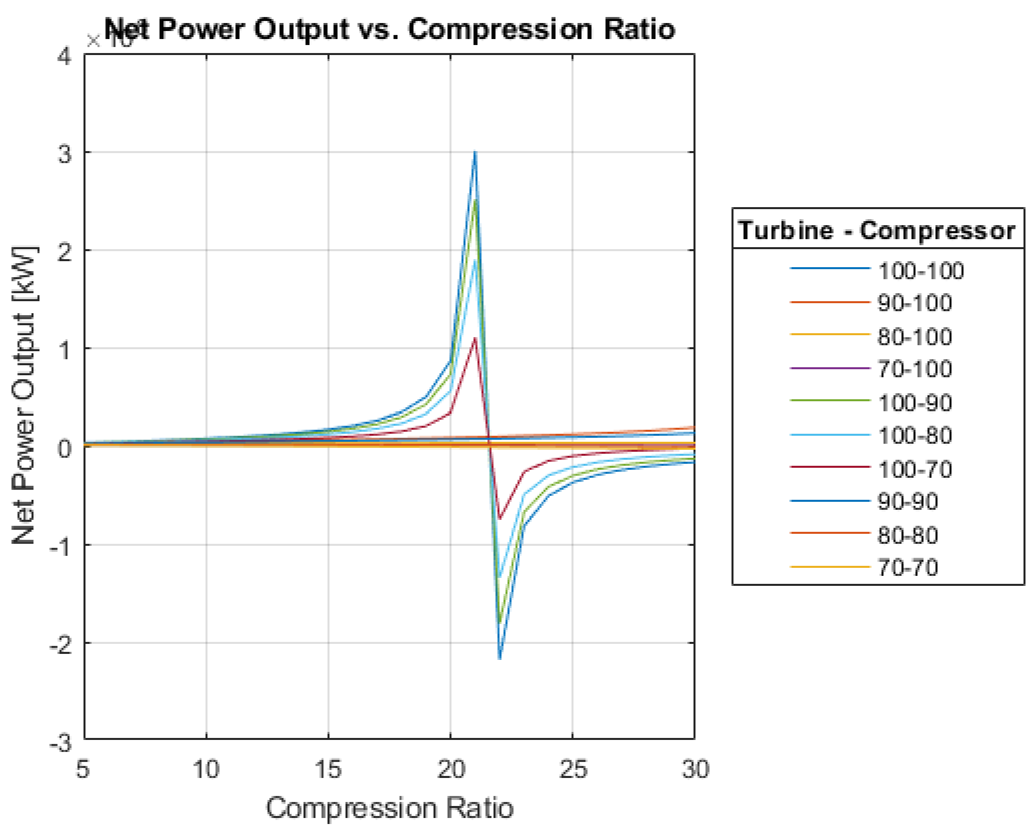
PART 1: RECOMMENDEDATION OF COMPONENT OMPTIMIZATION

```
% Clean up environment
clear;
close all;
clc;

% Set up constants (Working fluid of Brayton Cycle is Air)
k = 1.400;
Cp = 1.004;         % [kJ/kg-K]

% Set up known values
T1 = 300;           % [K]
T3 = 1200;          % [K]
T5 = 500;           % [K]

Q_L = 30*10^3;      % [kW]

% Set up table of compression ratios and iso-efficiencies
rp = 5:1:30;        % Compression Ratios from 5-30
n_turb = [1.00 .90  .80  .70  1.00 1.00 1.00 .90 .80 .70];
n_comp = [1.00 1.00 1.00 1.00 .90  .80  .70  .90 .80 .70];

% Set up empty vectors
n = zeros(length(n_turb),length(rp));
Wnet = zeros(length(n_turb),length(rp));

% Set up loops to cycle through variables
for z = 1:1:length(n_turb)
    for j = 1:1:length(rp)

        % State 1 - 2
        T2S = T1*rp(j)^((k-1)/k);
        T2 = (1/n_comp(z))*(T2S-T1) + T1;
        wC = Cp*(T2 - T1);

        % State 2 - 3
        qH = Cp*(T3-T2);

        % State 3 - 4
        T4S = T3*(1/rp(j))^((k-1)/k);
        T4 = T3 - n_turb(z)*(T3-T4S);
        wT = Cp*(T3-T4);

        % State 4 - 5
        qL = Cp*(T4-T5);

        % Efficiency of System + Net Power Output (nonspecific)
        %n(z,j) = 1-rp(j)^((1-k)/k);
```

```
        m = Q_L/qL;
        Wnet(z,j) = m*(wT-wC);

        QH = m*qH;
        n(z,j) = Wnet(z,j)/QH;

    end
end

% Find Optimal Compression Ratio for Each Combo
[mx,idx] = max(n,[],2);
opt_rps = rp(idx);

% Generate Plots
figure(1);
plot(rp,n)
grid on
title('Thermal Efficiency vs. Compression Ratio')
xlabel('Compression Ratio')
ylabel('Efficiency')
legend('100-100','90-100','80-100','70-100','100-90','100-80','100-70', ...
    '90-90','80-80','70-70',"Location","eastoutside");
title(legend,'Turbine - Compressor','FontSize',10)

figure(2);
plot(rp,Wnet)
grid on
title('Net Power Output vs. Compression Ratio')
xlabel('Compression Ratio')
ylabel('Net Power Output [kW]')
legend('100-100','90-100','80-100','70-100','100-90','100-80','100-70', ...
    '90-90','80-80','70-70',"Location","eastoutside");
title(legend,'Turbine - Compressor','FontSize',10)
```

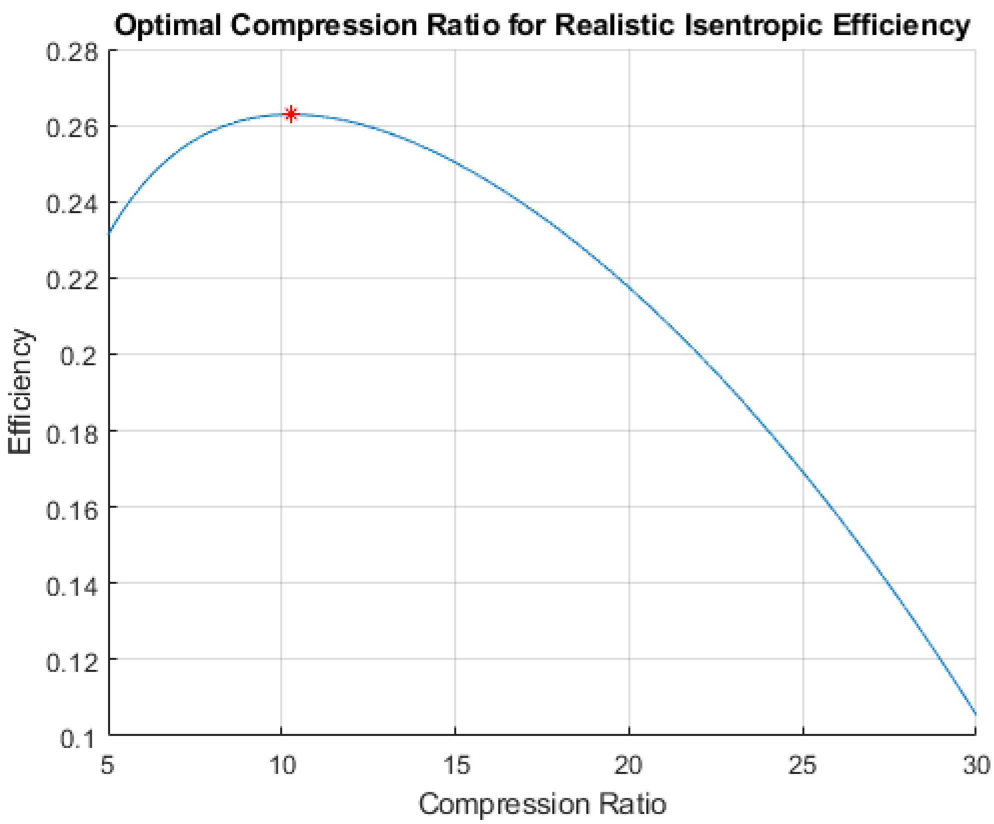

Thermal Efficiency vs. Compression Ratio
Efficiency
Compression Ratio
Turbine - Compressor
100-100
90-100
80-100
70-100
100-90
100-80
100-70
90-90
80-80
70-70


Net Power Output vs. Compression Ratio
Net Power Output [kW]
Compression Ratio
Turbine - Compressor
100-100
90-100
80-100
70-100
100-90
100-80
100-70
90-90
80-80
70-70

PART 2: OPTIMAL COMPRESSION RATIO

```
% Set up constants (Working fluid of Brayton Cycle is Air)
k = 1.400;
Cp = 1.004;         % [kJ/kg-K]

% Set up known values
T1 = 300;           % [K]
T3 = 1200;          % [K]
T5 = 500;           % [K]

Q_L = 30*10^3;      % [kW]

n_turb = 0.835;
n_comp = 0.835;
rp = 5:0.001:30;
n = zeros(1,length(rp));
Wnet = zeros(1,length(rp));
m = zeros(1,length(rp));

for j = 1:1:length(rp)

    % State 1 - 2
    T2S = T1*rp(j)^((k-1)/k);
    T2 = (1/n_comp)*(T2S-T1) + T1;
    wC = Cp*(T2 - T1);

    % State 2 - 3
    qH = Cp*(T3-T2);

    % State 3 - 4
    T4S = T3*(1/rp(j))^((k-1)/k);
    T4(j) = T3 - n_turb*(T3-T4S);
    wT = Cp*(T3-T4(j));

    % State 4 - 5
    qL = Cp*(T4(j)-T5);

    % Efficiency of System + Net Power Output (nonspecific)
    %n(z,j) = 1-rp(j)^((1-k)/k);

    m(j) = Q_L/qL;
    Wnet(j) = m(j)*(wT-wC);

    QH(j) = m(j)*qH;
    n(j) = Wnet(j)/QH(j);

end

[max, index] = max(n);
opt_rp = rp(index);
opt_m = m(index);
opt_Wnet = Wnet(index);
```

```
opt_QH = QH(index);
opt_n = max;
opt_T4 = T4(index);

figure;
hold on
grid on
plot(rp,n);
plot(opt_rp,max,'r*')
title('Optimal Compression Ratio for Realistic Isentropic Efficiency')
xlabel('Compression Ratio')
ylabel('Efficiency')
```



PART 3: CALCULATE COST

```
% Calculate Cost of Fuel (QH is used from PART 2)
heat_rate = 50*10^3;        % [kJ/kg]
unit_cost = 7.51/28.31;           % [$/L] or [$/kg] because they are
equivalent

% QH is in units of kJ/s --> QH/heat_rate gives kg/s burned
fuel_used = QH/heat_rate;                % [kg/s]

costPerTime = unit_cost*fuel_used;       % [$/kg * kg/s] = [$/s]
```

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Design Homework 3

```
clc;
clear;
close all;

% Set Up Knowns From Previous DHWs
T4 = 439.99;          % Celsius
T5 = 226.85;          % Celsius
T6 = 34.622;          % Celsius
T7 = 400;             % Celsius
Tsat = 179.8;         % Celsius

m_steam = 9.6202;
m_air = 140.168;

% Intermediate Heat Exchanger Temps and Heat Transfers were calculated by
% hand. Values found are listed below

% Superheater
T4B = 391.08;          % Celsius
Qs = 6731;             % kJ/s

% Evaporator
T4C = 225.6;           % Celsius
Qe = 23270.59;         % kJ/s

% Preheater
T5p = 225.53;           % Celsius
Qp = 9.427;            % kJ/s

% Efficiency of Whole Cycle is also found by hand

% Solve for the tube lengths

% SUPERHEATER

    % Constants
    do = 80E-3;
    di = 50E-3;
    N = 200;
    nh = 0.4;
    nc = 0.3;

% Steam
mu_steam = 19.1E-6;
Pr_steam = 1.52;
k_steam = 63.7E-3;

    % Reynolds Number
    dh = do-di;
    Re_steam = (dh*m_steam)/(mu_steam*N*(pi/4)*(do^2-di^2));
```

```
    % Nusselt Number
    Nu_steam = 0.023*Re_steam^(4/5)*Pr_steam^nh;

    % Heat Transfer Coefficient
    h_steam = (Nu_steam*k_steam)/dh;

% Air
mu_air = 338.8E-7;
Pr_air = 0.695;
k_air = 52.4E-3;

    % Reynolds Number
    Re_air = (4*m_air)/(N*pi*mu_air*di);

    % Nusselt Number
    Nu_air = 0.023*Re_air^(4/5)*Pr_air^nc;

    % Heat Transfer Coefficient
    h_air = (Nu_air*k_air)/di;

    % Using Log-Mean
    delT1 = T4 - T7;
    delT2 = T4B - Tsat;
    Tlm_s = (delT2-delT1)/log(delT2/delT1);

    U = ((1/h_air)+(1/h_steam))^-1;

    L_superheater = (Qs*10^3)/(N*pi*di*U*Tlm_s);


% Evaporator

% Water undergoes phase change, so h = inf, and goes to zero

% Air
mu_air = 298.84E-7;
Pr_air = 0.685;
k_air = 45.5E-3;

    % Reynolds Number
    Re_air = (4*m_air)/(N*pi*mu_air*di);

    % Nusselt Number
    Nu_air = 0.023*Re_air^(4/5)*Pr_air^nc;

    % Heat Transfer Coefficient
    h_air = (Nu_air*k_air)/di;

    % Using Log-Mean
    delT1 = T4B - Tsat;
    delT2 = T4C - Tsat;
    Tlm_e = (delT2-delT1)/log(delT2/delT1);
```

```
    U = ((1/h_air))^-1;

    L_evaporator = (Qe*10^3)/(N*pi*di*U*Tlm_e);

    qdp = Qe/(pi*di*N*L_evaporator);
% Preheater

d = 20E-3;
bafflespace = 0.2;
S_D = 1.25*d;
S_m = bafflespace*(S_D-d);

% Steam
mu_steam = 118E-6;
Pr_steam = 0.86;
k_steam = 642E-3;
cp_steam = 4.66;

    % Reynolds Number
    G_s = m_steam/S_m;
    De = (4*0.86*S_D^2-(pi/4)*d^2)/(pi*d);
    Re_steam = (De*G_s)/(N*mu_steam);

    % Nusselt Number
    Nu_steam = 0.36*Re_steam^(0.55)*Pr_steam^(1/3);

    % Heat Transfer Coefficient
    h_steam = (Nu_steam*k_steam)/d;

% Air
mu_air = 270.1E-7;
Pr_air = 0.683;
k_air = 40.7E-3;
cp_air = 1.030;

    % Reynolds Number
    G_s = m_air/S_m;
    De = (4*0.86*S_D^2-(pi/4)*d^2)/(pi*d);
    Re_air = (De*G_s)/(N*mu_air);

    % Nusselt Number
    Nu_steam = 0.36*Re_air^(0.55)*Pr_air^(1/3);

    % Heat Transfer Coefficient
    h_air = (Nu_air*k_air)/d;

    Up = ((1/h_air)+(1/h_steam))^-1;

% Log-mean

    delT1 = T5 - T6;
    delT2 = T4C - Tsat;
    Tlm_p = (delT2-delT1)/log(delT2/delT1);
```

```
    L_preheater = (Qp*10^3)/(pi*d*Up*Tlm_p);

% Critical Heat Flux

Cz = 0.131;
hfg = 1825;
rho_v = (1/0.0766);
rho_l = (1/(1.203E-3));
sigma = 31.6E-3;
g = 9.81;

qcrit = Cz*hfg*rho_v*((sigma*g*(rho_l-rho_v)/rho_v^2));
```

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